When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation

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Abstract In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, documents form interdependent networks, yet those relationships are typically represented as textual references rather than navigable structures. This paper argues that traceability is not only a property of documentation but an HCI problem about how document structure shapes navigation, and how navigation behavior reveals the structure practitioners are forced to work through. I introduce two design contributions that address this problem from complementary directions: an external traceability visibility prototype that makes document linkage navigable without modifying validated source files, and a behavioral navigation tracking prototype that operationalizes loopback behavior as an empirical indicator of document-structural friction. Two prototype-based studies with $N = 20$ GMP-experienced participants establish the empirical basis: Study 1 evaluated a graph-based Traceability Explorer over a synthetic corpus of 150 GMP records, identifying 31 missing CAPA links, 103 broken downstream links, and 21 lifecycle mismatch risks, with path reconstruction time reduced from approximately 7 minutes to approximately 4 minutes and all 20 participants completing target pathways successfully; Study 2 deployed a Cognitive Navigation Explorer during a standardized out-of-specification investigation scenario, observing loopback behavior across all 20 sessions (mean: approximately 5 loopbacks), with experienced practitioners averaging 2–3 loopbacks and less experienced participants averaging 7–10, while participants consistently attributed navigation difficulty to document structure rather than interface design. Three contributions follow: document traceability is introduced as a design problem specific to regulated environments; the no-modification constraint is identified as a novel design boundary for document navigation research; and making document relationships visible is argued to redistribute epistemic access by externalizing navigational structure that would otherwise remain implicit, unevenly distributed, and individually absorbed.

Keywords: document traceability, loopback behavior, GMP, pharmaceutical manufacturing, navigation friction, epistemic access, regulated environments, invisible work

Introduction

In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, documents do not exist in isolation. A single deviation investigation may require navigating across standard operating procedures, analytical method protocols, validation records, change control documents, and specification files — each of which references others through textual cross-references rather than navigable links. When these references are complete and current, traceability is possible. When they are missing, outdated, or structurally ambiguous, the burden of reconstructing pathways falls entirely on the practitioner. The practitioner becomes, in effect, the missing infrastructure between documents.

This paper argues that traceability is not only a property of documentation — whether references exist, whether audit trails can be reconstructed — but a condition of interaction. When document relationships are only partially visible, practitioners must navigate through incomplete structures, and their navigation behavior reflects the structure they are working through. A loopback — a return to a previously visited document — is not simply evidence of inefficient searching. It is evidence that the path forward was not yet legible: that the document system did not resolve the judgment demand on the first pass. In information foraging terms (Pirolli and Card, 1999), a loopback is a rational response to weak information scent, not a navigational error. This reframing has methodological consequences: loopback frequency is an empirical indicator of document-structural friction, not of individual reasoning quality or interface usability failure. It can be used to evaluate document systems from the outside, without deployment inside validated systems.

I report two prototype-based studies in GMP environments. Study 1 develops and evaluates a Traceability Explorer — an external graph-based visibility layer over a synthetic corpus of 150 GMP records — and demonstrates that making linkage structure navigable reduces path reconstruction burden. Study 2 develops and evaluates a Cognitive Navigation Explorer that records document transitions, visit frequency, and loopback events during a standardized out-of-specification investigation scenario, demonstrating that loopback frequency varies systematically with experience in ways that reflect document structure rather than individual reasoning quality. Neither prototype modifies validated source documents — a constraint constitutive of GMP-regulated environments, and one that defines a design boundary with no precedent in general document navigation research.

The contributions of this work are threefold. First, I introduce document traceability as an HCI problem specific to regulated environments, identifying the no-modification constraint as a novel design boundary for document navigation research. Second, I show that loopback analysis can function as an empirical method for surfacing document-structural friction without deployment inside validated systems — establishing a methodological tool for HCI research

in compliance- constrained domains where live system access is unavailable. Third, I offer a theoretical framing that connects document architecture to navigation behavior to accountability conditions: in GMP environments, whether a practitioner can reconstruct a defensible pathway depends not only on their expertise but on whether the document system makes its own structure legible, and the design gap between experienced and less experienced practitioners is itself an accountability finding.

Related Work

Documents as Organizational Action

The foundational premise of this study draws on Harper's (1998) ethnographic account of documentation practices at the IMF. Harper shows that documents are not neutral containers: their accessibility, their linkage to other documents, and the navigational demands they impose on users are constitutive of organizational action. When document relationships are visible and traversable, practitioners can reconstruct accountable pathways; when they are not, that reconstruction work falls to individuals. In GMP-regulated environments, this observation takes on regulatory force: the practitioner who manually bridges a broken traceability link is performing real work that the quality system does not measure, and producing an accountable pathway that the document system did not make available. The design question the present study addresses is how to make those relationships navigable from the outside, without modifying the documents that constitute them.

Articulation Work and the Human as Missing Infrastructure

Schmidt and Bannon (1992) introduce the concept of articulation work to describe the coordinative labor that makes other work possible — the work of aligning, scheduling, allocating, and monitoring that sustains collaborative activity without appearing in any formal record. In regulated document environments, cross-reference reconstruction is a form of articulation work: when linkage structure is incomplete or invisible, practitioners must manually bridge the gaps between documents, inferring relationships from partial cues and memory. This labor is not represented in any quality record, not measured by any quality metric, and not attributed to the document architecture that requires it. The practitioner who spends seven minutes reconstructing a traceability pathway before they can begin an out-of-specification investigation is performing articulation work that the quality system treats as part of the investigation itself. Study 1 demonstrates that an external traceability layer can absorb a portion of this articulation work into the design layer, redistributing it from individual, invisible effort to shared, navigable structure.

Information Foraging and Loopback as Structural Signal

Pirolli and Card (1999) develop information foraging theory to explain how people navigate information environments: users follow information scent — cues that indicate the likely value of information patches — and move between patches in ways that maximize gain relative to cost. A loopback is a rational response to weak or misleading scent: when the path forward does not provide sufficient signal, the practitioner returns to a prior position to reorient. Applied to GMP document navigation, loopbacks indicate that the document system did not provide sufficient structural cues to support forward navigation on the first pass. The practitioner who returns to the ELISA Test Method SOP three times during an OOS investigation is not confused; they are navigating a document system whose cross-reference structure does not resolve the judgment demand the investigation requires. Loopback frequency is therefore a readout of document-structural friction: a measure of how many times the system failed to provide sufficient scent to sustain forward progress. Study 2 demonstrates that this measure is both sensitive — varying systematically with experience level — and attributable: participants consistently described their loopbacks as responses to document structure rather than interface design.

Situated Action and Invisible Work

Suchman's (1987) account of situated action establishes that navigation through a document system is shaped by the immediate conditions of execution — what is visible, accessible, and interpretable at the moment a decision must be made. The practitioner navigating an OOS investigation is not following a pre-specified plan; they are constructing a pathway through documents whose relationships are only partially legible, adapting to what each document makes available. When that adaptation requires loopbacks — returns to prior documents to reestablish context that later documents did not provide — it is a situational response to document structure, not a reasoning failure.

Star and Strauss (1999) show that work performed in the gaps of formal systems tends to become invisible. In regulated document environments, the manual reconstruction of broken traceability pathways is precisely this kind of invisible work: not represented in investigation records, not measured by quality metrics, and not attributed to the document architecture that produces it. The experience asymmetry in loopback frequency — in which less experienced practitioners perform more traceability reconstruction labor because they have not yet internalized the document system's implicit map — is a specific instance of the epistemic access inequality that **?** describes: structural conditions restrict who can effectively navigate the knowledge system, not by competence but by accumulated implicit knowledge the system does not externalize.

Graph-Based Document Analysis

Network analysis provides formal methods for examining relational document structures. **?** establish centrality measures — in-degree, betweenness, PageRank — as tools for identifying structurally important nodes in directed graphs. Applied to GMP document networks, graph analysis can identify which documents are structurally central to traceability pathways, which linkage gaps are most consequential for pathway reconstruction, and which documents are most likely to become navigation bottlenecks when their cross-references are incomplete. Study 1 applies this methodology to a synthetic GMP corpus, using directed graph representation to identify missing links, broken downstream references, and lifecycle mismatches that are not visible from within individual document records.

Prototype-Based Evaluation

The methodological approach builds on Nielsen (1994)'s demonstration that think-aloud evaluation with small domain-specific samples reliably surfaces usability and friction patterns. Hutchinson et al. (2003) establish the logic of the technology probe: a prototype used simultaneously as a design intervention and as an analytic instrument, making it possible to study how structural conditions shape behavior rather than simply to evaluate whether a given interface is usable. This dual function is central to Study 2 in particular: the Cognitive Navigation Explorer was not deployed to evaluate its own usability but to make visible the navigational conditions of GMP document work — the loopback patterns that document-structural friction produces — in a way that live system observation within a validated GMP environment would not permit.

Method

Participants

Both studies were conducted with the same participant pool of $N = 20$, recruited from the same pharmaceutical quality team, with GMP work experience and direct familiarity with deviation or out-of-specification investigation workflows. Experience levels ranged from limited ($<$2 years, $n = 3$), intermediate ($n = 14$), to extensive ($\geq$5 years, $n = 3$). The same-sample design eliminates between-participant variance in domain knowledge as a confound; recruiting from a single expert team prioritized internal consistency over sample diversity, at the cost of generalizability acknowledged in the limitations.

Synthetic Document Corpora

Both studies used synthetic document corpora constructed in Python to reflect the structural and terminological characteristics of real GMP documentation without reproducing proprietary content. This construction approach reflects a domain constraint: GMP quality records are protected under confidentiality obligations that preclude their use in external research without facility authorization. Synthetic generation from structural parameters is the standard approach for research in regulated manufacturing contexts where live data access is unavailable, and the structural validity of the findings does not depend on the specific content of the documents but on the navigational and linkage patterns they instantiate.

Study 1: Traceability Explorer

The Study 1 corpus represented an interdependent network of 150 GMP records modeled as a directed graph using Python and NetworkX. Each record was assigned a type (deviation, CAPA, change control, SOP, or specification), a lifecycle status, and cross-references to related records. Graph analysis identified three categories of linkage problem: 31 missing CAPA links (deviations without associated corrective action records), 103 broken downstream links (records referencing superseded or nonexistent documents), and 21 lifecycle mismatches (closed deviations with open CAPAs, or active SOPs referenced by records that postdate their supersession).

Based on this analysis, I developed an external Traceability Explorer as a Streamlit web application organized into three tabs (Figures 1–3): a **Chain Viewer** presenting the four-record documentation chain for any selected record with missing and broken links flagged inline; a **Risk Summary** surfacing traceability risks at the dataset level with sortable detail tables; and a **Record Explorer** allowing filtering across the full corpus by record type, status, and product. The no-modification constraint was operationalized throughout: the prototype reads from the graph representation of the corpus but does not modify, version, or interact with any source document.

Participants completed three task-based traceability scenarios, each requiring reconstruction of a specific documentation pathway: identifying the CAPA associated with a given deviation, tracing the change control history of a specified SOP, and identifying all records affected by a lifecycle mismatch. Post-task measures included task completion time, path completion accuracy, document discovery rate, ease of use, and adoption intent.

Study 2: Cognitive Navigation Explorer

Participants navigated across seven synthetic GMP documents during a standardized out-of-specification investigation scenario. The seven documents included the deviation record, the

relevant test method SOP, the instrument qualification record, the analyst training record, the environmental monitoring specification, the change control record for a recent instrument modification, and the CAPA associated with a prior related deviation. These documents were connected by cross-references of varying completeness: some links were explicit and current; others were implicit, outdated, or absent.

The Cognitive Navigation Explorer (Figures 4–6) recorded document-to-document transitions with action type (forward navigation, cross-reference jump, or loopback), visit count, time spent per document, and the full navigation sequence. A loopback was operationally defined as any return to a previously visited document where the intervening path had visited at least one other document. This definition excludes immediate re-reads (which may reflect careful attention rather than navigation failure) while capturing the pattern of interest: returns that indicate the practitioner found insufficient information to sustain forward progress and needed to reorient from a prior position.

Findings

Study 1: Traceability Visibility

Path reconstruction time reduced from approximately 7 minutes to approximately 4 minutes — a 30–60% reduction that was consistent across task types and experience levels. All 20 participants completed target pathway reconstruction, rated ease of use at 5.0 out of 5, and expressed adoption intent. These quantitative results are consistent with the prototype's design intent, but the qualitative responses provide the more consequential finding: participants described the prototype not primarily as a time-saving tool but as a legibility tool — one that made relationships visible that were previously invisible, rather than merely making existing navigation faster.

Participant responses clustered around two themes. The first was discovery: participants found linkage gaps and lifecycle mismatches they had not known existed. "I had no idea we had 31 missing CAPA links. That's not a number I could have generated from the system I currently use." The second was the handover use case, which participants described as the most immediately valuable application: "every time someone leaves the team, the first three months of their replacement is basically archaeology. This would change that." A third participant described the prototype in terms of what it externalizes: "right now, the map of how these documents connect exists only in the heads of the people who've been here long enough to build it. This makes it a system property instead of a person property."

This third account articulates the study's theoretical claim in practitioner terms: the prototype does not create new information but externalizes structural knowledge that currently exists only as accumulated implicit expertise. Making it a system property redistributes the epistemic access

that currently depends on tenure.

Study 2: Behavioral Navigation Tracking

Loopback behavior was observed across all 20 participant sessions, with a mean of approximately 5 loopbacks per session. The distribution by experience level was systematic: experienced practitioners ($\geq$5 years) averaged 2–3 loopbacks; intermediate participants averaged 4–6; less experienced participants averaged 7–10. This stratification was consistent across navigation tasks and document types.

The most common loopback target was the ELISA Test Method SOP, which participants returned to an average of 2.1 times per session. Post-navigation discussion clarified the mechanism: the SOP contained cross-references to the instrument qualification record and the environmental monitoring specification, but these references were textual rather than navigable, and the referenced documents were not consistently current. Participants returned to the SOP not because they had failed to read it carefully on the first visit, but because later documents raised questions whose answers they expected the SOP to contain — and because the SOP's cross-reference structure did not resolve those questions on the first pass.

Less experienced participants described their loopbacks in structural terms: "I kept going back to the ELISA SOP because I wasn't sure if I'd missed something — there were so many cross-references and I couldn't tell which ones actually mattered." A second participant described the same dynamic differently: "I kept thinking the answer was in the SOP. But the SOP kept pointing me to documents that pointed me back to the SOP. I was going in circles because the circle was in the documents, not in my head." Experienced practitioners described fewer loopbacks and explained the difference in terms of internalized knowledge: "I know which cross-references to follow and which ones are just there for compliance. You learn that over time. But it's not written down anywhere."

This experience asymmetry is the central finding of Study 2. It is not a reasoning gap: experienced and less experienced practitioners were using the same navigation strategy, responding rationally to the same information scent. The difference is structural: experienced practitioners have internalized a map of which cross-references are navigable and which are nominal, which documents are likely to resolve a given judgment demand and which will redirect it. That map is not currently a system property; it is a person property, unevenly distributed by tenure and not externalized by the document system.

Discussion

Traceability as an Interaction Problem

Traceability in regulated environments is not only a documentation property — a question of whether the required links exist and can be audited — but a condition of interaction: a question of whether practitioners can navigate those links under realistic task conditions. The distinction matters for design. If traceability is a documentation property, the appropriate intervention is records management: ensuring that links are created, maintained, and auditable. If traceability is an interaction condition, the appropriate intervention is different: ensuring that the relationships represented in documentation are navigable from the outside, by practitioners who need to reconstruct pathways without modifying the documents that constitute them.

Study 1 demonstrates that making linkage structure externally navigable changes the conditions of pathway reconstruction. The 30–60% reduction in path reconstruction time is significant; the qualitative shift from individual archaeological labor to shared structural legibility is more so. Practitioners described the prototype not as a faster way to do the same thing but as a way to do something the current system does not support: navigate the document network as a structure rather than as a collection of individually accessed records.

Loopback as Method

The contribution of Study 2 is not only substantive but methodological. Loopback analysis provides a way to evaluate document systems from the outside — without deployment inside validated systems, without access to live quality records, and without requiring practitioners to report their navigation difficulties directly. Loopback frequency is a behavioral indicator that is sensitive to document- structural friction, interpretable in information foraging terms (Pirolli and Card, 1999), and attributable by participants to document structure rather than interface design.

This methodological contribution has implications for HCI research in compliance-constrained domains more broadly. Many regulated environments restrict researchers from observing live system use or accessing production data. Cognitive navigation prototypes that capture loopback behavior in realistic task scenarios provide a way to study navigation friction empirically, without requiring the deployment inside validated systems that GMP regulation prohibits. The operational definition used here — loopback as any return to a previously visited document where the intervening path visited at least one other document — is specific enough to be reproducible and general enough to apply across document environments. Future work should examine whether loopback frequency correlates with downstream accountability outcomes, and whether reducing loopback frequency through structural visibility interventions produces measurable changes in investigation quality.

Epistemic Access in Regulated Work

The experience asymmetry documented in Study 2 is not a training gap. Training would address a deficit in practitioner knowledge; the asymmetry documented here reflects a deficit in system structure. Experienced practitioners navigate the document network more efficiently not because they reason better but because they have internalized a map of the network that the system does not provide: which cross-references are navigable, which documents are likely to contain the information a given query demands, which structural gaps require workarounds and which are genuine absences.

This asymmetry is a form of epistemic injustice in the sense that **?** describes: structural conditions restrict who can effectively navigate the knowledge system, not by competence but by accumulated implicit knowledge the system does not externalize. Making linkage structure visible through the Traceability Explorer externalizes this map — transferring knowledge from the implicit possession of experienced practitioners to the explicit navigability of all. This redistribution is not merely an efficiency gain; it is an equity intervention. The practitioner who requires seven loopbacks to reconstruct an OOS investigation pathway is not performing a worse investigation than the practitioner who requires two. They are performing a more expensive one, absorbing costs that the document system produced and the quality system does not measure.

The No-Modification Constraint as Design Boundary

Both studies operate under a design constraint that has no precedent in general document navigation research: interventions must operate externally over document representations without modifying validated source files. This constraint is not peripheral; it defines what counts as a permissible intervention in GMP contexts and distinguishes Regulatory HCI from general information architecture. In non-regulated environments, navigation problems can often be addressed by restructuring the information itself — adding explicit links, redesigning information architecture, creating navigational overlays that modify source content. In GMP environments, none of these are permissible; the validated status of quality records prohibits modification without regulatory consequence.

The no-modification constraint is therefore both a design limitation and a research contribution: it identifies a class of design problems — compliance- constrained navigation — that has not been adequately addressed in the HCI literature, and for which the standard toolkit of information architecture and navigation design does not apply. External graph-based traceability layers and behavioral navigation tracking prototypes are two responses to this constraint; future work should develop the design vocabulary for this space more fully.

Limitations and Future Work

The use of synthetic corpora and a single-team participant pool are the primary limitations. The synthetic corpora instantiate the structural patterns of interest — missing links, broken references, lifecycle mismatches — but may underrepresent the terminological and organizational complexity of live GMP document networks. The single-team participant pool produces consistent baseline knowledge at the cost of generalizability: findings may not transfer to quality teams with different experience distributions, document system architectures, or regulatory frameworks.

Future work should deploy both prototypes in live document environments, using the external graph representation approach to analyze real document networks without accessing protected content. The handover use case identified by participants in Study 1 is particularly promising: personnel transitions create a natural experimental context in which the value of externalized navigational knowledge is acute and measurable. Future work should also examine whether loopback frequency correlates with downstream accountability outcomes — whether practitioners with higher loopback rates produce investigation conclusions that differ systematically from those with lower rates — and whether reducing loopback frequency through structural visibility interventions changes the quality of those conclusions.

Conclusion

This paper introduced document traceability as a condition of interaction in regulated environments — a structural determinant of who can navigate defensible pathways, at what cost, and under what conditions. Study 1 demonstrated that a graph-based Traceability Explorer reduced path reconstruction time by 30–60%, with all 20 participants completing target pathways and expressing adoption intent, and with qualitative responses framing the prototype as an epistemic redistribution — externalizing navigational knowledge that currently exists only as individually accumulated implicit expertise. Study 2 demonstrated loopback behavior across all 20 sessions, with experience-stratified variation reflecting document structure rather than individual reasoning quality, and with participants consistently attributing their navigation difficulty to the document system's failure to externalize its own structural logic.

Three contributions follow. First, document traceability is introduced as an HCI problem specific to regulated environments, where the no-modification constraint creates a design boundary that prohibits the standard toolkit of information architecture and navigation design, and requires external observability layers that make document relationships navigable without touching the documents themselves. Second, loopback analysis is established as a methodological tool for studying document-structural friction in compliance-constrained domains, providing a

behavioral indicator that is sensitive, interpretable, and collectable without deployment inside validated systems. Third, the experience asymmetry in loopback frequency is identified as a design finding rather than a training gap: the navigational support that experienced practitioners have internalized is not currently a system property, and making it one redistributes epistemic access in ways that are equitable, measurable, and achievable within compliance constraints.

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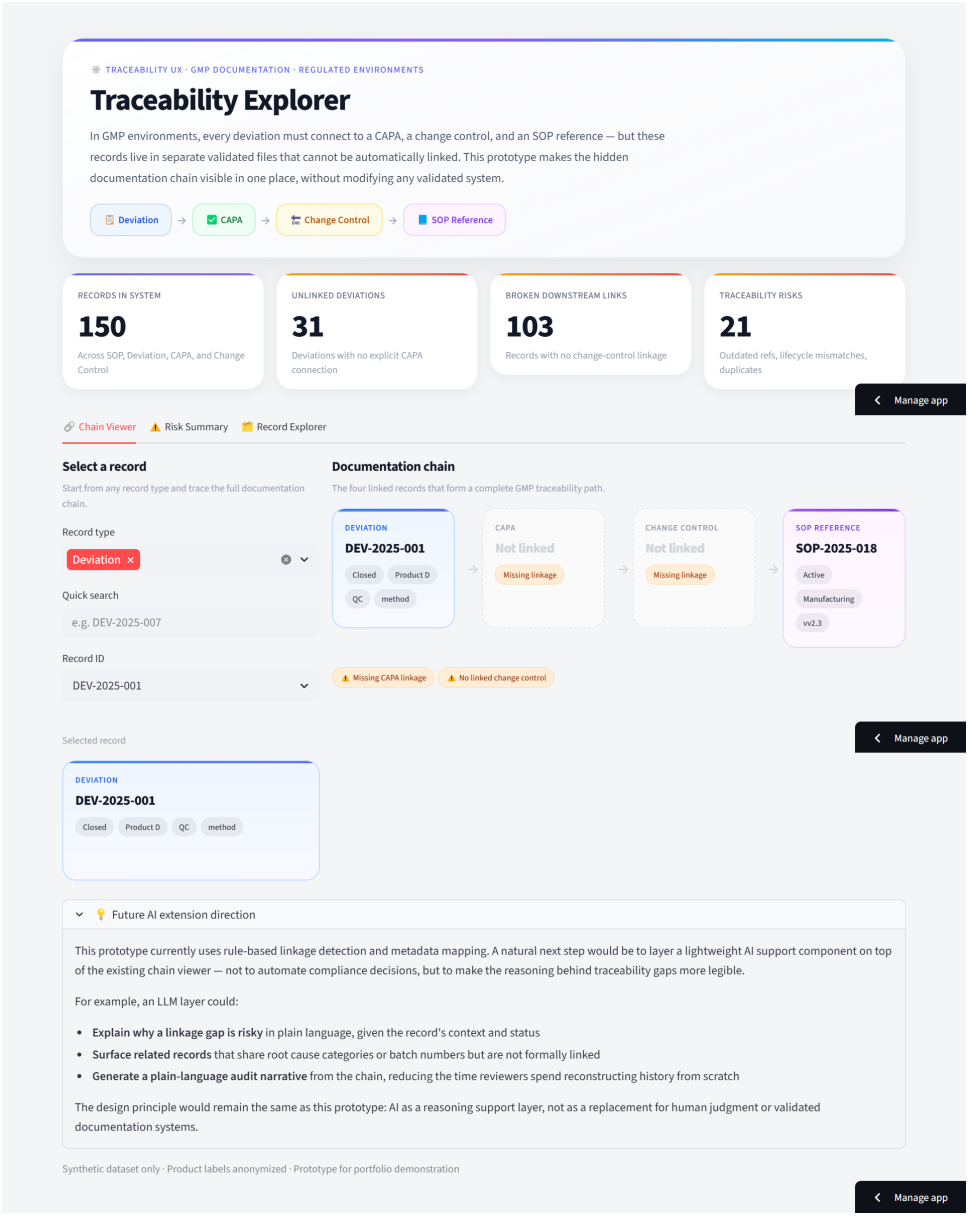


Figure 1: Traceability Explorer: Chain Viewer tab, showing the documentation chain for DEV-2025-001. CAPA and Change Control nodes are marked “Not linked,” while SOP-2025-018 is resolved as active and current.

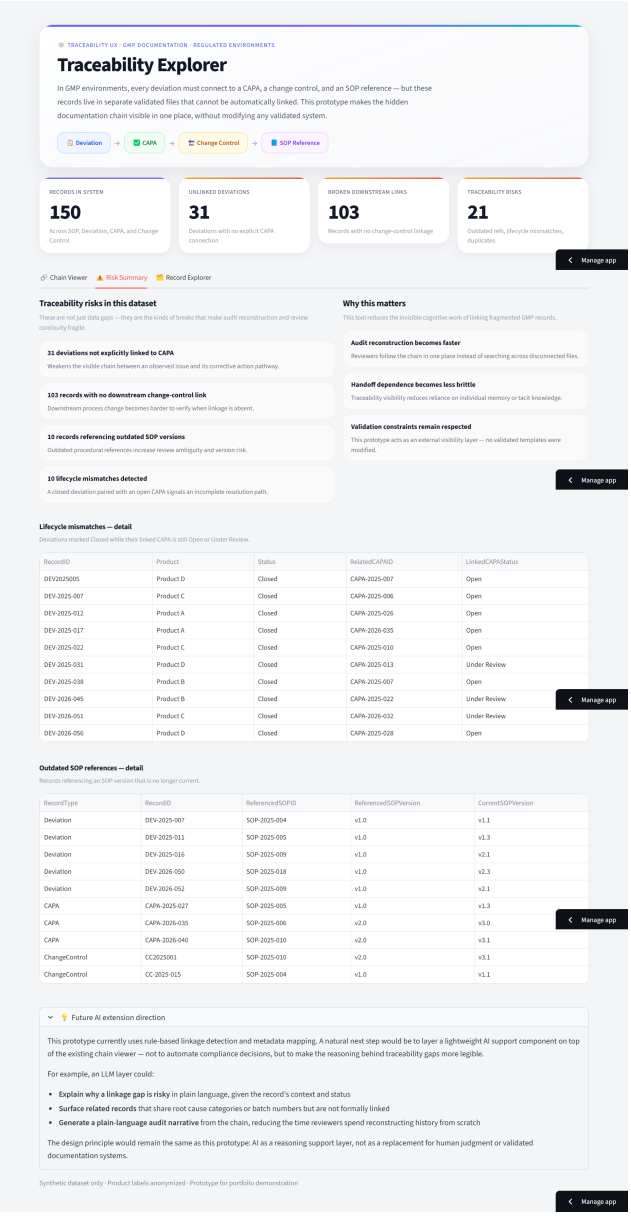


Figure 2: Traceability Explorer: Risk Summary tab, showing four risk categories. Lifecycle mismatches detail table shows 10 closed deviations whose linked CAPAs remain open. Outdated SOP references table shows 10 records referencing superseded versions.

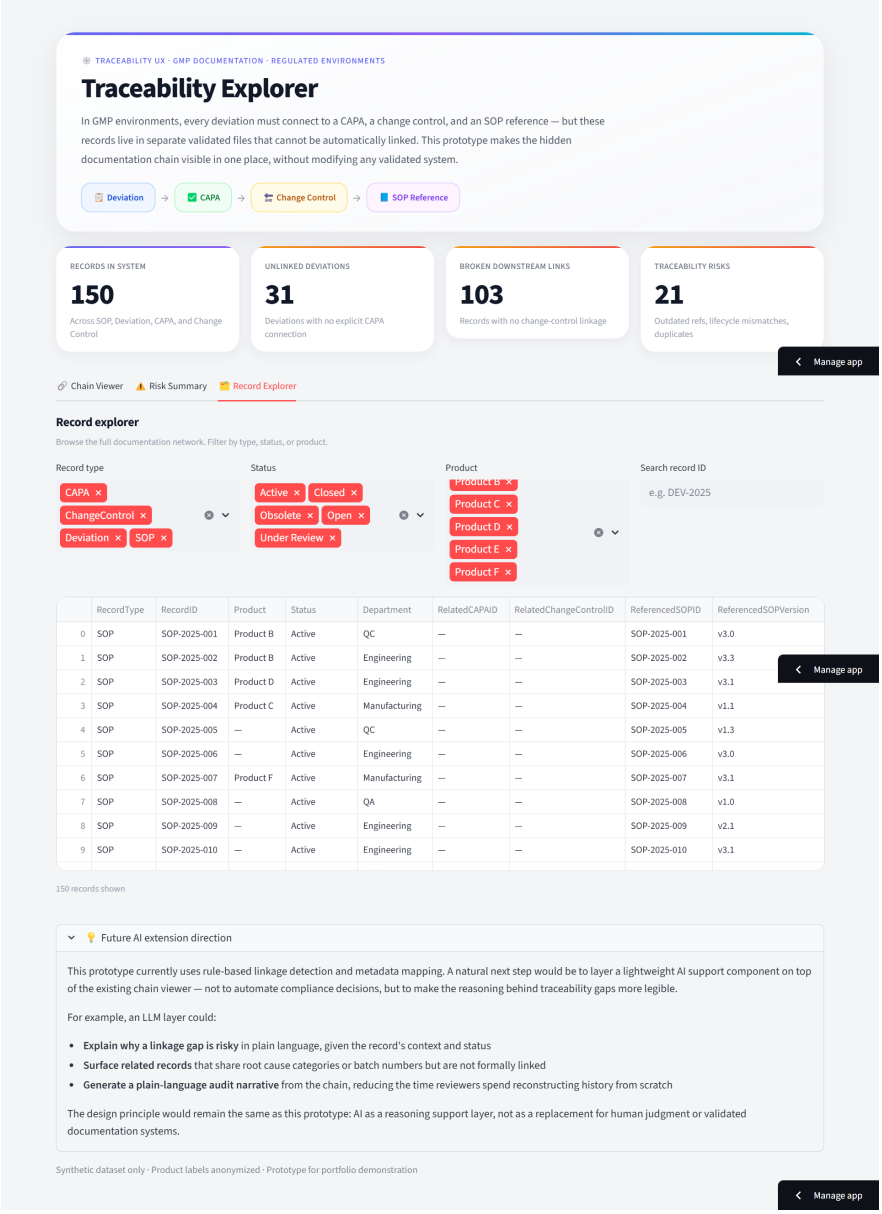


Figure 3: Traceability Explorer: Record Explorer tab, showing the full 150-record corpus with filters for record type, status, and product. Missing CAPA IDs and change control IDs are visible as empty cells.

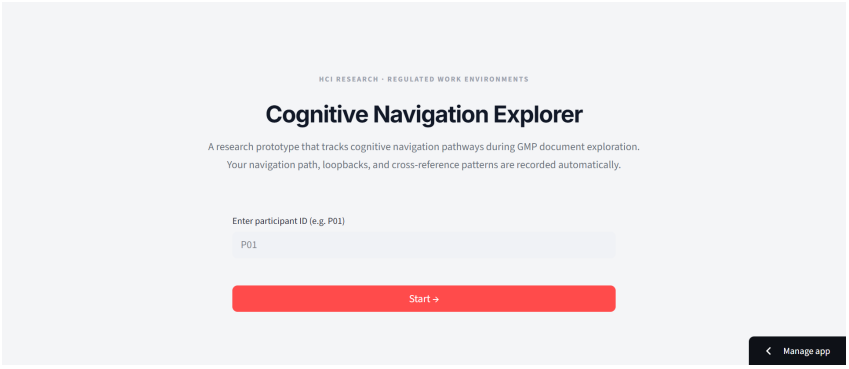


Figure 4: Cognitive Navigation Explorer: landing screen showing the study description and participant ID entry.

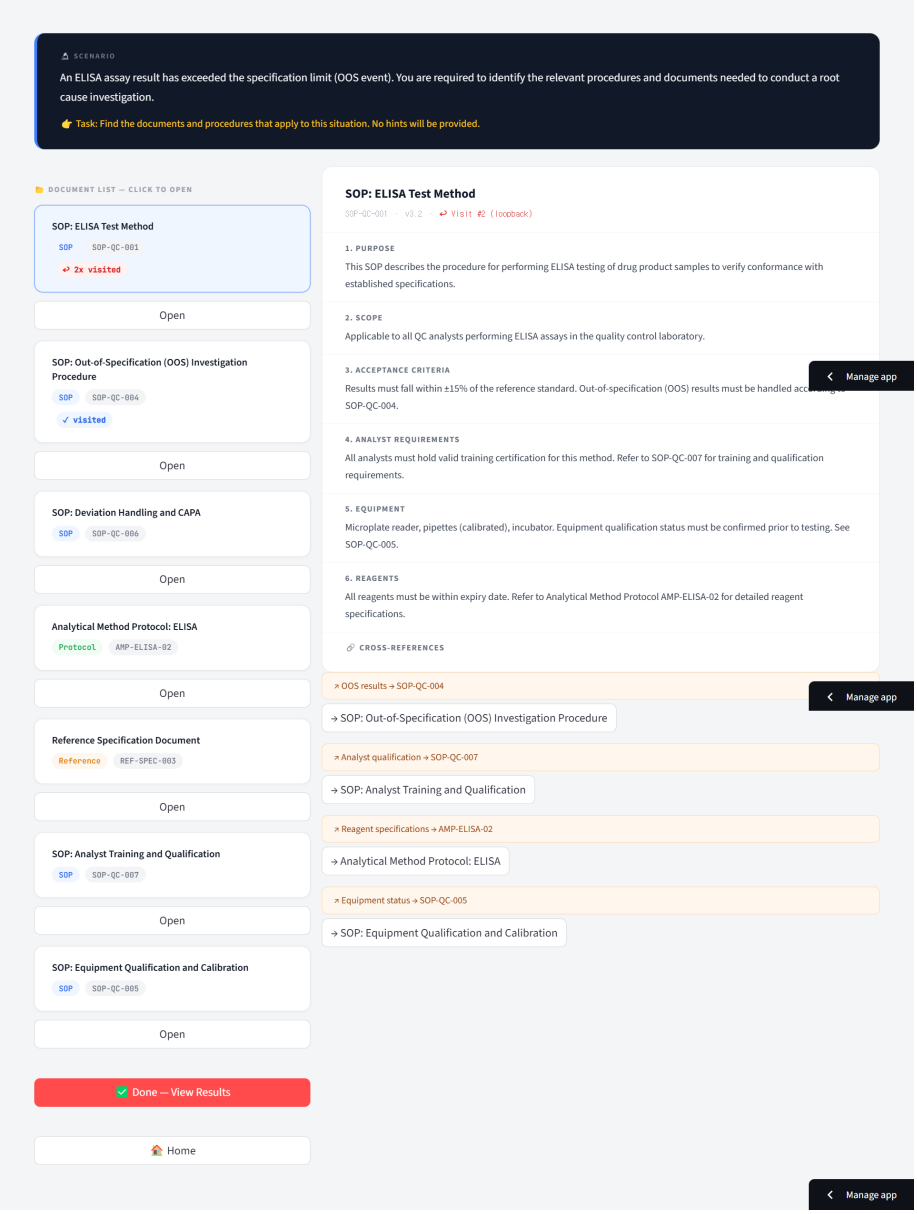


Figure 5: Cognitive Navigation Explorer: OOS investigation scenario in progress. The ELISA Test Method SOP is open at Visit #2 (loopback), indicated by the “2x visited” badge and the “Visit #2 (loopback)” label in the document header.

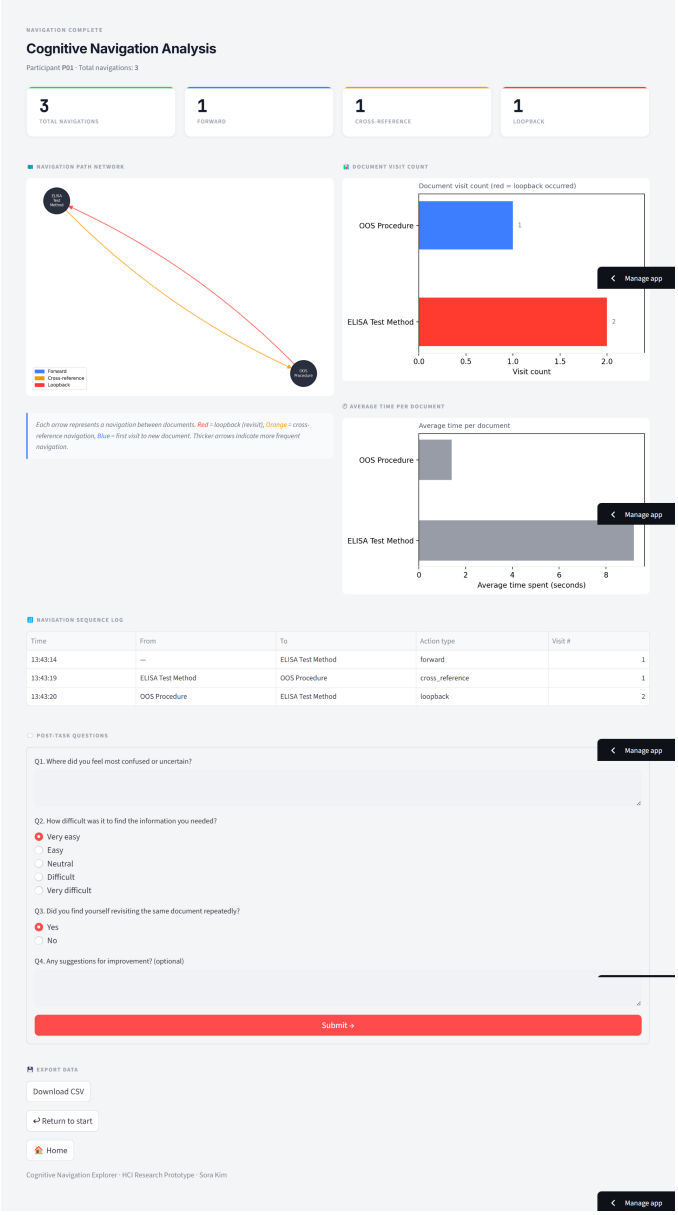


Figure 6: Cognitive Navigation Analysis: results dashboard for Participant P01, showing navigation path network and document visit counts. Red bars indicate loopback occurrence.